

1.2.9★ How many sets of four consecutive positive integers are there such that the product of the four integers is less than 100,000? **Hints:** 123, 104

1.3.4★ At the Gooddog Obedience School, dogs can learn to do three tricks: sit, stay, and roll over. Of the dogs at the school:

50 dogs can sit	17 dogs can sit and stay
29 dogs can stay	12 dogs can stay and roll over
34 dogs can roll over	18 dogs can sit and roll over
9 dogs can do all three	9 dogs can do none

(b) How many dogs can do exactly 2 tricks? **Hints:** 131

1.28 In the upcoming election, voters can choose:

- Abe or Barb for President;
 - Carlos, Darla, or Eve for Vice-President;
 - Fred, Gina, or Hal for Secretary.
- (c) How many ways can the ballot be completed if the voter may choose not to vote in some (or all) of the elections?

1.29 How many positive integers less than 500 can be written as the sum of two positive perfect cubes?
Hints: 35, 65

Problem 2.6: How many three-digit numbers are not multiples of 7?

Problem 2.7: The Smith family has 4 sons and 3 daughters. In how many ways can they be seated in a row of 7 chairs such that at least 2 boys are next to each other?

Problem 2.10: In how many ways can we pick a group of 3 different numbers from the group

$$1, 2, 3, \dots, 500$$

such that one number is the average of the other two? (The order in which we pick the numbers does not matter.)

Problem 2.14: Our math club has 20 members and 3 officers: President, Vice President, and Treasurer. However, one member, Ali, hates another member, Brenda. How many ways can we fill the offices if Ali refuses to serve as an officer if Brenda is also an officer?

Problem 2.15: We have the same club (20 members and 3 officers), but this time, Ali has a huge crush on Brenda, and won't be an officer unless she is one too. Brenda is unaware of Ali's affections, and doesn't care if he is an officer or not; she's perfectly happy to be an officer even if Ali isn't one. In how many ways can the club choose its officers?

2.5.5 Our club has 25 members, and wishes to pick a president, secretary, and treasurer. In how many ways can we choose the officers, if:

- (c) individual members are allowed to hold 2, but not all 3, offices? **Hints: 161**

2.19 How many triples (a, b, c) of even positive integers satisfy $a^3 + b^2 + c \leq 50$?

2.23 How many 4-digit numbers have the second digit even and the fourth digit at least twice the second digit?

2.29 The n members of a committee are numbered 1 through n . One of the members is designated as the "Grand Pooh-Bah." The n members sit in a row of n chairs, but no member with a number greater than the Grand Pooh-Bah may sit in the seat to the immediate right of the Grand Pooh-Bah. Suppose that the Grand Pooh-Bah is member number p , where $1 \leq p \leq n$. Find a formula, in terms of n and p , for the number of ways for the committee to sit. **Hints:** 220 (Source: MATHCOUNTS)

2.31 A palindrome is a number that reads the same forwards and backwards, such as 12321.

(d)★ Suppose we form all k -digit palindromes that consist only of 8's and 9's such that there is at least one of each digit. What is the smallest value of k such that there are at least 2004 numbers in the list? **Hints:** 46

2.32 Let S be a cube. Compute the number of planes which pass through at least three vertices of S . **Hints:** 152, 68

3.4.3 A Senate committee has 5 Democrats and 5 Republicans. In how many ways can they sit around a circular table:

(c) if each member sits next to two members of the other party? **Hints:** 194

3.26★ A regular tetrahedron is a triangular pyramid whose faces are all equilateral triangles. How many distinguishable ways can we paint the four faces of a regular tetrahedron with red, blue, green, and orange paint such that no two faces are the same color? **Hints:** 166

4.2.3 Our water polo team has 15 members. I want to choose a starting lineup consisting of 7 players, one of whom will be the goalie (the other six positions are interchangeable). In how many ways can I choose my starting lineup? **Hints:** 88

4.20 The Aviation Club at Rocco's school has 12 members (including Rocco). They need to choose a 3-person Helicopter Committee and a 4-person Glider Committee. Students can serve on either or both committees, but Rocco refuses to serve on both—he will only serve on one or the other (or neither). In how many ways can both committees be chosen?

4.22 There are 5 different pairs of gloves, where left and right are distinguishable. Select 4 from the 10 gloves.

(b) How many ways are there to select 4 such that some 2 of the 4 make a pair?

4.23* In poker, a 5-card hand is called a *three of a kind* if there are three cards of one rank and two other cards which are not the same rank as each other or as the other three cards. How many 5-card hands are three of a kind? (Assume we are using a standard 52-card deck with 13 ranks in each of 4 suits.)
Hints: 133, 214

5.3.5★ We call a number a *mountain number* if its middle digit is larger than any other digit. For example, 284 is a mountain number. How many 3-digit mountain numbers are there? **Hints:** 182, 12

5.4.3 How many ways are there to put 2 balls in n boxes (where $n \geq 2$) if

(d) the balls are not distinguishable and neither are the boxes?

5.21 How many positive integers less than 1000 are there such that the digits are not in strictly increasing order from left to right? (For example, the digits of 124 are in strictly increasing order, but the digits of 88 and 946 are not.) **Hints:** 208

5.25★ An integer is called *snakelike* if its decimal representation $a_1a_2a_3\cdots a_k$ satisfies $a_i < a_{i+1}$ if i is odd and $a_i > a_{i+1}$ if i is even. How many snakelike integers between 1000 and 9999 have four distinct digits? **Hints:** 41, 147 (Source: AIME)

5.26★ Mack the bug starts at the point $(0,0,0)$ at noon and each minute moves one unit in either the positive x -direction, the positive y -direction, or the positive z -direction. Thus, after 1 minute he could be at $(1,0,0)$, $(0,1,0)$, or $(0,0,1)$; after 2 minutes, he could be at $(2,0,0)$, $(0,2,0)$, $(0,0,2)$, $(1,1,0)$, $(1,0,1)$, or $(0,1,1)$. How many different paths could he take to $(3,5,2)$? **Hints:** 136

Problem 6.4: Nine parallel lines in a plane intersect a set of n parallel lines that go in another direction. The lines form a total of 360 parallelograms, many of which overlap each other. Find n . (Source: ARML)

Problem 6.6: In the grid shown in Figure 6.2, how many paths are there from A to B using only steps up or to the right? Note that a path cannot traverse over a missing edge!

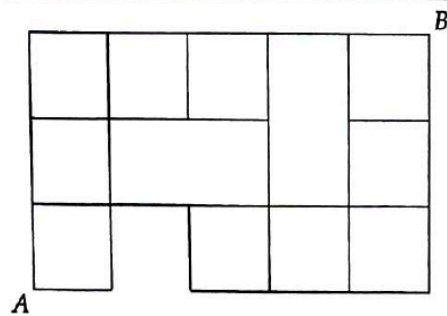


Figure 6.2: Grid for Problem 6.6

6.11 I have tiled my square bathroom wall with congruent square tiles. All the tiles are red, except those along the two diagonals, which are all blue (i.e. the corners are blue and all the tiles along the diagonals between each pair of opposite corners are blue). If I used 121 blue tiles, how many red ones did I use? **Hints:** 89 (Source: AMC)

6.12 A *lattice point* is a point with integer coordinates such as $(2,3)$. In how many ways can we pick 3 lattice points such that both coordinates of each point are positive integers less than 5, and the three points form a triangle? **Hints:** 192, 13

6.15 In this problem, we are constructing 12-letter words with four X's, four Y's, and four Z's.

(b) How many such words have no X's in the first 4 letters and no Y's in the next 4 letters?

6.16 For any positive integer n , let $O(n)$ be the sum of the odd digits of n and $E(n)$ be the sum of the even digits of n . Let

$$x = O(1) + O(2) + O(3) + \cdots + O(100)$$

and

$$y = E(1) + E(2) + E(3) + \cdots + E(100).$$

Find x and y . (Source: AMC)

6.19 An 8×8 checkerboard has alternating black and white squares, as shown to the right. How many distinct rectangles, with sides on the grid lines of the checkerboard and containing at least 4 black squares, can be drawn on the checkerboard?
Hints: 206

